

Literature-Implied Obstructions for the Complemented-Subspace Problem in L_1

Math discovery packet

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1 Status

This is a lightweight *literature-implied partial answer*, not a full solution of the source question.

The source paper is Dirk Werner, *Nigel Kalton's work in isometrical Banach space theory*, arXiv:1103.3153. In Section 5, “Operators on L_1 ”, immediately after Theorem 5.2, Werner writes that it is still open whether a complemented infinite-dimensional subspace of L_1 must be isomorphic to L_1 or ℓ_1 .

The full problem remains open in the checked literature. For example, arXiv:1811.06571 calls it the famous problem whether every infinite-dimensional complemented subspace of L_1 is isomorphic to ℓ_1 or L_1 , and arXiv:2310.02196 describes the separable L_1 -space CSP as a long-standing open problem.

2 Literature-Implied Subcases

The main supporting paper is

D. de Hevia, G. Martinez-Cervantes, A. Salguero-Alarcon, and P. Tradacete,

A negative solution to the complemented subspace problem in Banach lattices, arXiv:2310.02196.

Its Corollary 2.3 states that if X is a complemented subspace of an L_1 -space and X is isomorphic to a Banach lattice, then X is isomorphic to an L_1 -space. The proof combines the Lindenstrauss–Rosenthal fact that complemented subspaces of L_1 -spaces are $\mathcal{L}_1$ -spaces with Proposition 2.1 of the same paper, which says that a Banach lattice that is an $\mathcal{L}_1$ -space is lattice isomorphic to an L_1 -space.

Consequently, in the separable $L_1[0, 1]$ setting of Werner’s question, the Banach-lattice subcase reduces to the standard separable L_1 -space classification: $L_1[0, 1]$, ℓ_1 , or a finite-dimensional ℓ_1^n . Since the source question is infinite-dimensional, this gives:

Any counterexample must not be isomorphic to a Banach lattice.

There is a second classical obstruction. Lindenstrauss and Pelczynski’s Theorem 6.1 on $\mathcal{L}_1$ -spaces says that the only $\mathcal{L}_1$ -space with an unconditional basis is ℓ_1 . This theorem is quoted, for example, in arXiv:1912.08449 and applied in arXiv:1905.13188 to complemented subspaces of L_1 . Since a complemented subspace of L_1 is an $\mathcal{L}_1$ -space, one gets:

A complemented subspace of L_1 with an unconditional basis is isomorphic to ℓ_1 .

Thus any infinite-dimensional counterexample to Werner’s question must be a non-lattice $\mathcal{L}_1$ -space and must have no unconditional basis.

3 Failed Route

The counterexample in arXiv:2310.02196 to the Banach-lattice CSP does not appear to produce a counterexample for L_1 by duality. The paper notes that the dual of the Plebanek–Salguero space PS is a 1-complemented subspace of $C(K_B)^*$, hence is linearly isometric to $\ell_1(\Gamma)$ for some set Γ . Therefore the adjoint side falls into the known atomic L_1 alternative rather than producing a new complemented subspace of $L_1[0, 1]$.

4 Search Evidence and Scope

- Run indexes were searched for arXiv:1103.3153, the source title, Kalton/L1/complemented-subspace keywords, and close variants. No prior durable packet for this exact target was found.
- Local source checks included arXiv:1811.06571, arXiv:1905.13188, arXiv:1912.08449, and arXiv:2310.02196.
- External arXiv/web checks on 2026-06-21 found no later full resolution of the complemented-subspace problem for $L_1[0, 1]$.

This packet should be read as a structural narrowing of the search space, not as a solution of Werner’s problem. A full counterexample, if it exists, must evade both the Banach-lattice and unconditional-basis mechanisms above.

5 References

- D. Werner, *Nigel Kalton’s work in isometrical Banach space theory*, arXiv:1103.3153.
- D. de Hevia, G. Martinez-Cervantes, A. Salguero-Alarcon, and P. Tradacete, *A negative solution to the complemented subspace problem in Banach lattices*, arXiv:2310.02196.
- J. Lindenstrauss and A. Pelczynski, *Absolutely summing operators in $\mathcal{L}_p$ -spaces and their applications*, Studia Math. 29 (1968), 275–326.
- F. Albiac, J. L. Ansorena, and P. Wojtaszczyk, *On certain subspaces of ℓ_p for $0 < p \leq 1$ and their applications to conditional quasi-greedy bases in p -Banach spaces*, arXiv:1912.08449.
- M. Novotny, *Some Remarks on Schauder Bases in Lipschitz Free Spaces*, arXiv:1905.13188.
- W. B. Johnson, G. Schechtman, and T. Schlumprecht, *Ideals in $L(L_1)$* , arXiv:1811.06571.